

Supplemental Online Content

Combes A, Saura O, Nessler N, et al; LEVOECMO Trial Group; International ECMO Network (ECMONet). Levosimendan to facilitate weaning from ECMO in patients with severe cardiogenic shock: the LEVOECMO randomized clinical trial. *JAMA*. Published online December 1, 2025. doi:10.1001/jama.2025.19843

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This supplemental material has been provided by the authors to give readers additional information about their work.

eAppendix. Supplementary Data

Patients

Exclusion criteria were: age <18 years; pregnant or lactating women; initiation of VA-ECMO >48 h; resuscitation >30 minutes (cumulative low-flow time) in the 48 hours before ECMO; irreversible neurological pathology; end-stage cardiomyopathy with no hope of LV function recovery; mechanical complication of myocardial infarction; aortic regurgitation > grade II; VA-ECMO for pulmonary embolism; VA-ECMO for cardiotoxic drug intoxication; ECMO after left-ventricle assist device implantation; VA-ECMO in heart transplant patients; patient moribund on the day of randomization; SAPS II >90; liver cirrhosis (Child B or C) and other severe hepatic insufficiency; chronic renal failure requiring hemodialysis; known hypersensitivity to levosimendan; history of torsades de pointes in the 30 days prior to inclusion; history of epilepsy; individuals under guardianship; or permanently legally incompetent adults; participation to another interventional study; patient with a weight over 180 kg; known hypersensitivity to polyvitamin CERNEVIT®. As patients couldn't provide written consent at inclusion, it was obtained from a close relative or surrogate. If unavailable, emergency consent allowed randomization without it. The patient and relative/legal representative were informed as soon as possible and gave written consent for continuation.

eTable 1. Type of Consent

	Overall N = 146 ¹	Levosimendan N = 69 ¹	Placebo N = 77 ¹
Emergency consent, no. (%)			
Patient's signed continued informed consent	113/146 (77.4%)	53/69 (76.8%)	60/77 (77.9%)
Patient unable to consent at the end of follow-up	4/146 (2.7%)	1/69 (1.4%)	3/77 (3.9%)
Death before ability to consent	29/146 (19.9%)	15/69 (21.7%)	14/77 (18.2%)
Declined consent	0/146 (0.0%)	0/69 (0.0%)	0/77 (0.0%)
	Overall N = 59 ¹	Levosimendan N = 32 ¹	Placebo N = 27 ¹
Consent from a close relative or surrogate, no. (%)			
Patient's signed continued informed consent	35/59 (59.3%)	20/32 (62.5%)	15/27 (55.6%)
Patient unable to consent at the end of follow-up	3/59 (5.1%)	1/32 (3.1%)	2/27 (7.4%)
Death before ability to consent	21/59 (35.6%)	11/32 (34.4%)	10/27 (37.0%)
Declined consent	0/59 (0.0%)	0/32 (0.0%)	0/27 (0.0%)
¹ n/N (%)			

eTable 2. Cause-Specific Analysis

Event	csHR (95% CI) ^a	<i>P</i> value
Successful ECMO weaning	1.15 (0.82 – 1.62)	0.42
ECMO weaning failure	0.91 (0.46 – 1.82)	0.79
Death before ECMO weaning	1.57 (0.72 – 3.44)	0.26

^a: Cause-specific hazard ratio (csHR) for the primary endpoint, estimated using the cause-specific Cox proportional hazards model and adjusted for cardiogenic shock etiology.

eTable 3. Secondary End Points

Outcomes/events^a	Levosimendan N = 101	Placebo N = 104	Median or Risk Difference, (95% CI)^b	Relative difference (95% CI)^c
Number of organ-failure-free days at Day 30, median (IQR)	12 (0, 21)	14 (0, 23)	-2 (-10 to 4)	
Number of days with organ failure, median (IQR)	9 (6, 17)	12 (6, 18)	-3 (-5 to 2)	
Number of catecholamine-free days at Day 30, median (IQR)	18 (0, 23)	18 (5, 25)	0 (-4 to 6)	
Catecholamine duration, days, median (IQR)	9 (5, 13)	10 (5, 16)	-1 (-4 to 2)	
No of days alive and free from cardiovascular failure within 7 days d, median (IQR)	0 (0, 2)	0 (0, 2)	0 (0 to 0)	
No of days alive and free from MV by day 30, median (IQR)	6 (3, 11)	6 (3, 13)	-0 (-3 to 2)	
Mechanical ventilation duration by day 30, days, median (IQR)	21 (0, 27)	22 (2, 27)	-0 (-7 to 4)	
Kidney replacement therapy duration by day 30, days, median (IQR)	30 (0, 30)	30 (8, 30)	0 (-5 to 1)	
No of days alive and free from KRT by day 30, median (IQR)	0 (0, 2)	0 (0, 5)	0 (0 to 0)	
Left ventricular ejection fraction, median (IQR), by day 30	45 (30, 50) (n = 100)	40 (25, 50)	5 (-0 to 10)	
Major adverse cardiovascular events during first 30 days, no. (%)	35 (34.7%)	36 (34.6%)	0.0 (-12.6 to 13.6)	RR, 1.00 (0.69 to 1.46)
Cardiac transplant by day 30, no. (%)	0 (0.0%)	5 (4.8%)	-4.8 (-9.6 to -1.0)	-
Escalation to permanent left ventricular assist device by day 30, no. (%)	4 (4.0%)	4 (3.8%)	0.1 (-4.8 to 5.1)	RR, 1.03 (0.26 to 4.01)
Stroke by day 30, no. (%)	7 (6.9%)	6 (5.8%)	1.2 (-5.7 to 8.0)	RR, 1.20 (0.42 to 3.45)
Dialysis by day 30, no. (%)	29 (28.7%)	38 (36.5%)	-7.8 (-21.4 to 4.7)	RR, 0.79 (0.53 to 1.17)

Rehospitalization for heart failure by day 30, no. (%)	0 (0.0%)	1 (1.0%)	-1.0 (-2.9 to 0.0)	-
Major adverse cardiovascular events during first 60 days, no. (%)	36 (35.6%)	39 (37.5%)	-1.9 (-15.4 to 11.7)	RR, 0.95 (0.66 to 1.36)
Cardiac transplant by day 60, no. (%)	0 (0.0%)	5 (4.8%)	-4.8 (-9.6 to -1.0)	-
Escalation to permanent left ventricular assist device by day 60, no. (%)	4 (4.0%)	5 (4.8%)	-0.8 (-6.7 to 5.0)	RR, 0.82 (0.23 to 2.98)
Stroke by day 60, no. (%)	7 (6.9%)	6 (5.8%)	1.2 (-5.7 to 8.0)	RR, 1.20 (0.42 to 3.45)
Dialysis by day 60, no. (%)	-	-		
Rehospitalization for heart failure by day 60, no. (%)	0 (0.0%)	1 (1.0%)	-1.0 (-2.9 to 0.0)	-

Abbreviations: IQR, Interquartile range; Q1, first quartile (25th percentile); Q3, third quartile (75th percentile); RR, Relative Risk; CI, Confidence Interval; ECMO, extracorporeal membrane oxygenation, ECMO, extracorporeal membrane oxygenation; MV, mechanical ventilation, KRT, kidney replacement therapy.

^a: Summary measure within each group: median (interquartile range) for continuous variables, number and percentage (n (%)) for categorical variables. For patients alive at day 30, left ventricular ejection fraction (LVEF) was assessed at day 30 and for those who died before day 30, the last available LVEF measurement prior to death was used. In the presence of missing data, the number of observations is reported as (n = ...) for continuous variables and as n/N (%) for categorical variables. Being free from cardiovascular failure was defined as a cardiovascular component of the SOFA score < 2.

^b: Absolute difference between groups (difference in medians for continuous variables or in percentages for categorical variables), with a 95% confidence interval estimated by non-parametric bootstrap resampling.

^c: Relative measure of effect: expressed as relative risk (95% CI) for binary endpoints. The relative risk and its 95% confidence interval are not estimated when one of the groups has zero events.

p-values were not reported for secondary outcomes because of multiple testing concerns.

eTable 4. Daily Doses of Dobutamine, Epinephrine and Norepinephrine From Randomization to Day 7

Drug	Day	Placebo, n; Median (IQR)	Levosimendan, n; Median (IQR)
Dobutamine	0	88 ; 9.44 (5.00 - 11.85)	81 ; 8.43 (5.00 - 10.10)
	1	89 ; 9.70 (5.00 - 12.00)	83 ; 8.41 (5.00 - 11.25)
	2	88 ; 8.56 (5.00 - 10.33)	80 ; 7.33 (4.98 - 10.00)
	3	89 ; 5.02 (4.22 - 9.84)	69 ; 6.67 (4.97 - 10.00)
	4	82 ; 5.00 (3.00 - 8.08)	61 ; 5.00 (4.46 - 8.00)
	5	74 ; 5.00 (2.70 - 8.70)	58 ; 4.98 (3.55 - 7.35)
	6	55 ; 5.00 (3.41 - 8.13)	48 ; 5.00 (3.98 - 7.04)
	7	55 ; 5.20 (4.25 - 9.13)	43 ; 5.00 (4.00 - 7.83)
Epinephrine	0	-	2 ; 0.02 (0.01 - 0.02)
	1	-	2 ; 0.02 (0.01 - 0.02)
	5	-	1 ; 0.10 (0.10 - 0.10)
	7	-	1 ; 2.00 (2.00 - 2.00)
Norepinephrine	0	76 ; 0.32 (0.10 - 0.76)	72 ; 0.40 (0.18 - 0.71)
	1	80 ; 0.40 (0.17 - 0.94)	80 ; 0.63 (0.30 - 1.02)
	2	70 ; 0.30 (0.10 - 0.59)	80 ; 0.52 (0.21 - 0.98)
	3	58 ; 0.26 (0.11 - 0.73)	61 ; 0.40 (0.20 - 0.83)
	4	50 ; 0.26 (0.10 - 0.90)	48 ; 0.38 (0.18 - 0.66)
	5	46 ; 0.30 (0.10 - 0.78)	51 ; 0.25 (0.15 - 0.48)
	6	38 ; 0.32 (0.20 - 0.69)	46 ; 0.26 (0.16 - 0.49)
	7	38 ; 0.20 (0.10 - 0.56)	42 ; 0.35 (0.14 - 0.58)

eTable 5. Vasopressor or Inotrope Exposure and Treatment Duration From Randomization to Day 30

Drug	Outcome	Levosimendan	Placebo
Dobutamine	Exposure	(93/101) (92.1%)	(96/104) (92.3%)
	Duration	6 [4-10]	7 [4.75-13]
Epinephrine	Exposure	(8/101) (7.9%)	(5/104) (4.8%)
	Duration	1 [1-1]	1 [1-2]
Norepinephrine	Exposure	(93/101) (92.1%)	(98/104) (94.2%)
	Duration	7 [4-11]	5 [3-12]
Other vasoactive drugs ^a	Exposure	(14/101) (13.9%)	(12/104) (11.5%)
	Duration	2.5 [1.25-3]	2 [1-3.25]

^a Other drugs included vasopressin, isoprenaline and milrinone.

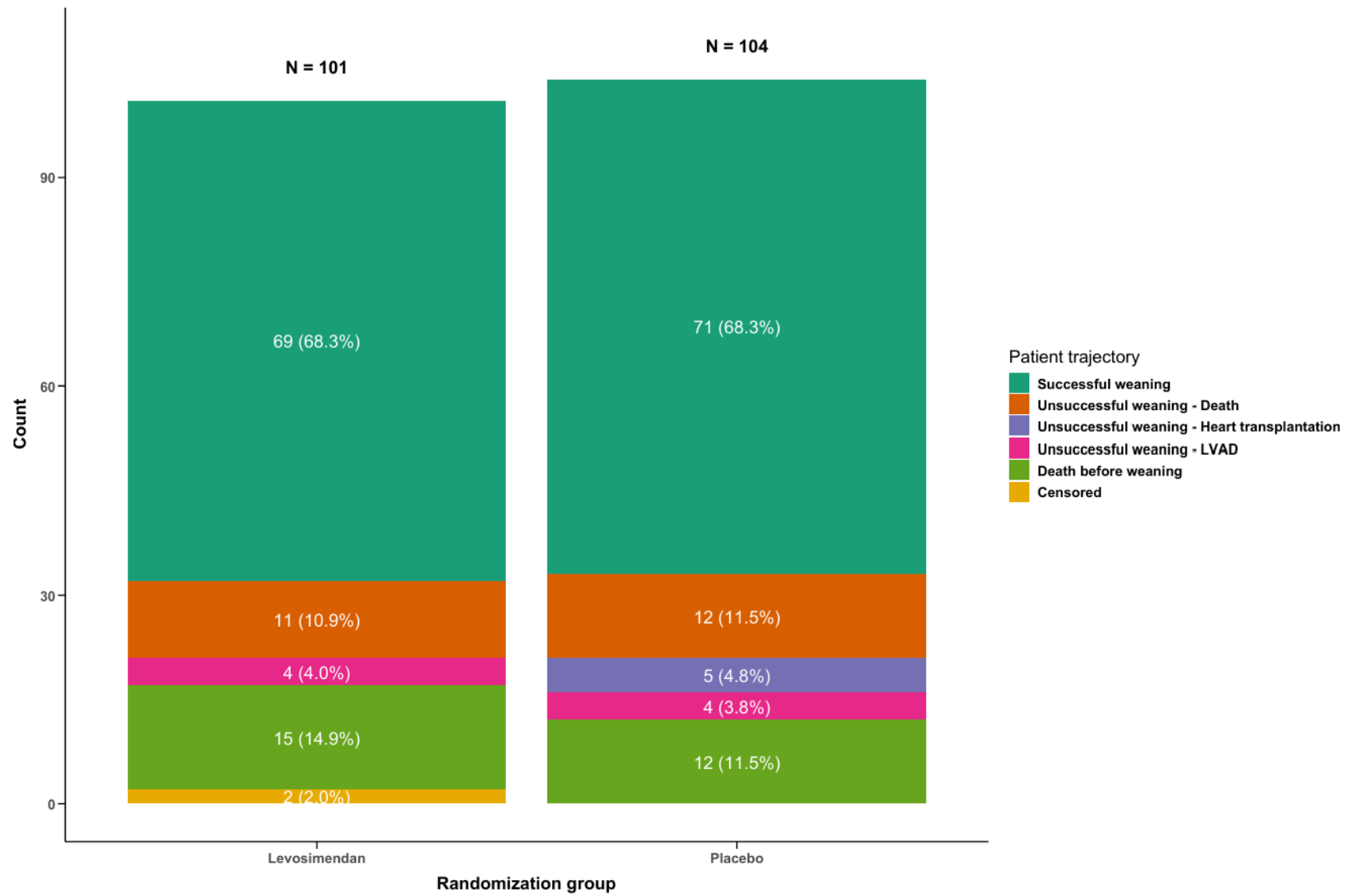
eTable 6. Other Causes of Cardiogenic Shock at Randomization

	Levosimendan N = 21 ^a	Placebo N = 21 ^a
Other causes of cardiogenic shock, no. (%)		
Decompensated dilated cardiomyopathy	3/21 (14.3%)	9/21 (42.9%)
Acute cardiac arrhythmia	6/21 (28.6%)	6/21 (28.6%)
Cardiac arrest prior to ECMO	5/21 (23.8%)	1/21 (4.8%)
Takotsubo cardiomyopathy	3/21 (14.3%)	1/21 (4.8%)
Acute valvular disease	1/21 (4.8%)	3/21 (14.3%)
Miscellaneous conditions ^b	3/21 (14.3%)	1/21 (4.8%)

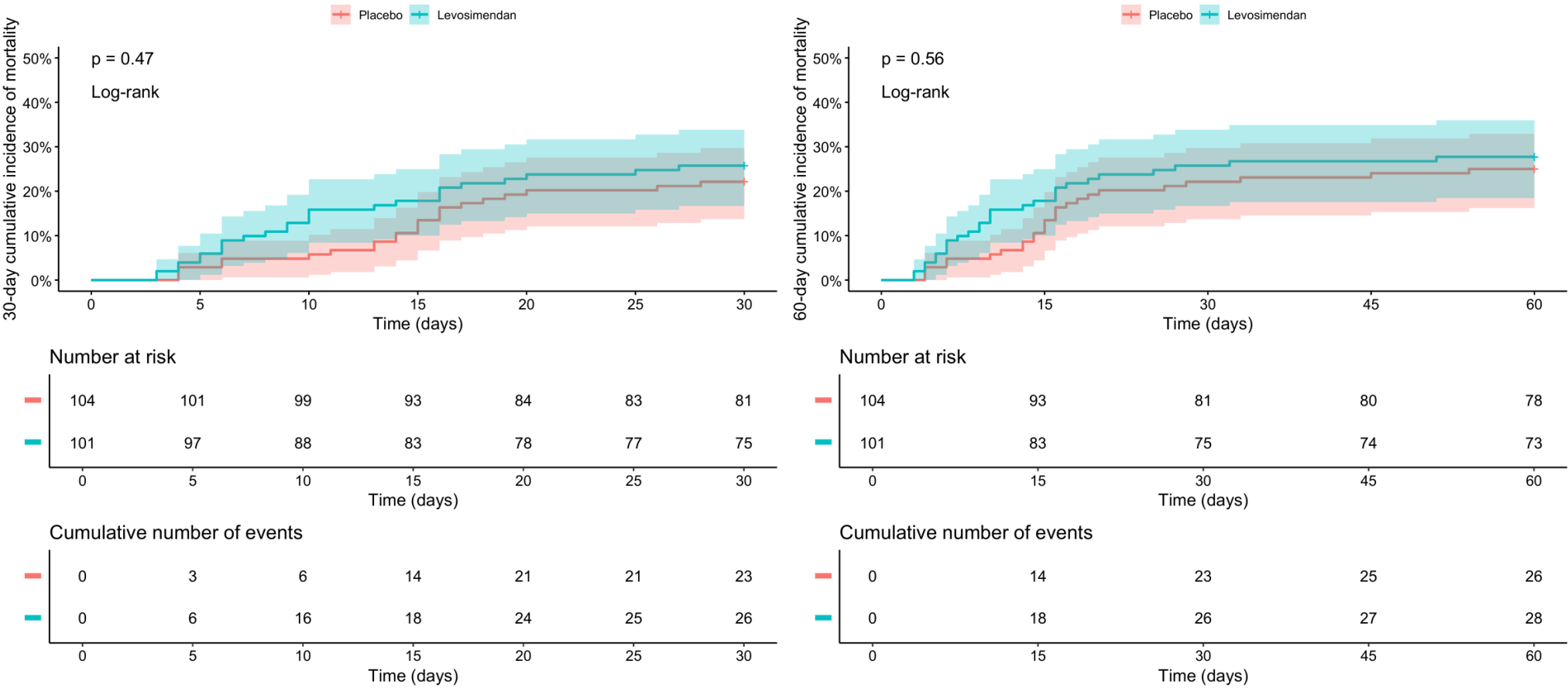
^a n/N (%)

^b Miscellaneous conditions such as sepsis, late complications of congenital heart disease, and thyrotoxic cardiomyopathy.

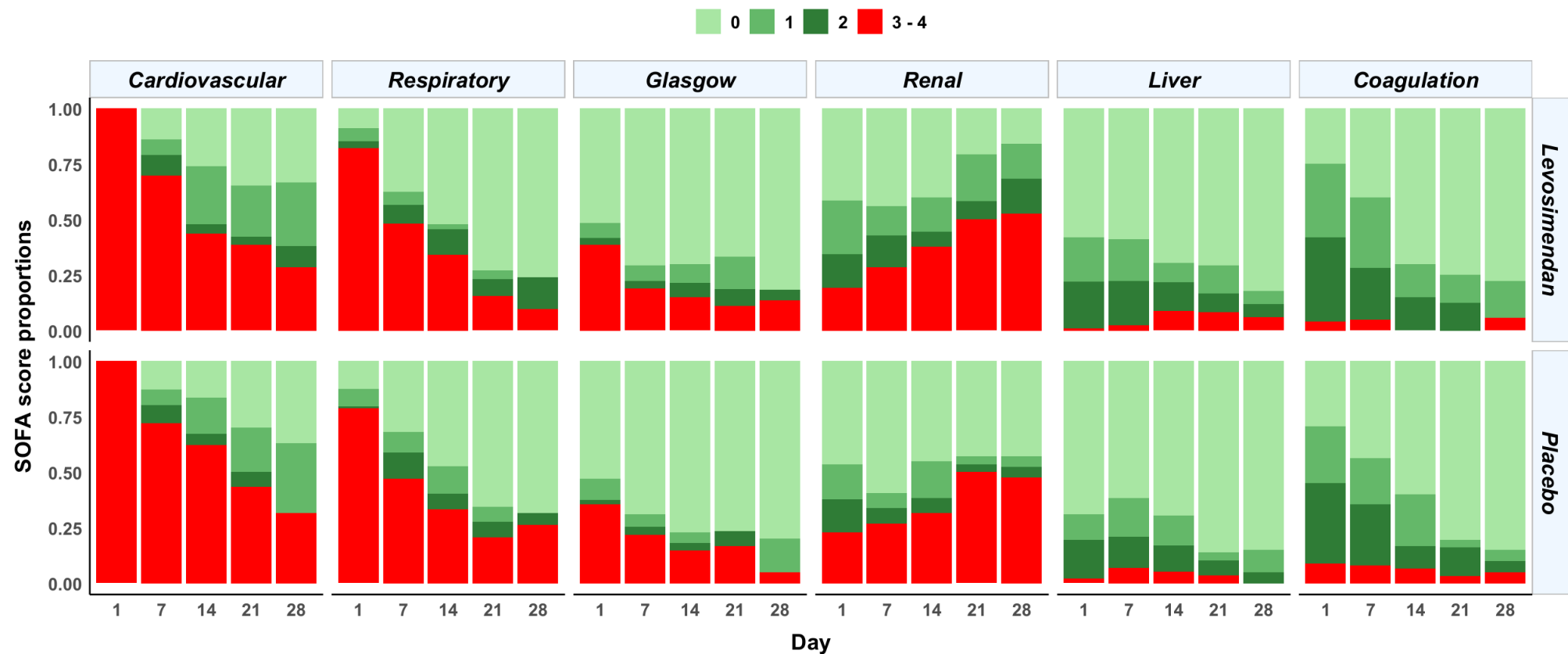
eFigure 1. Patients' Trajectory



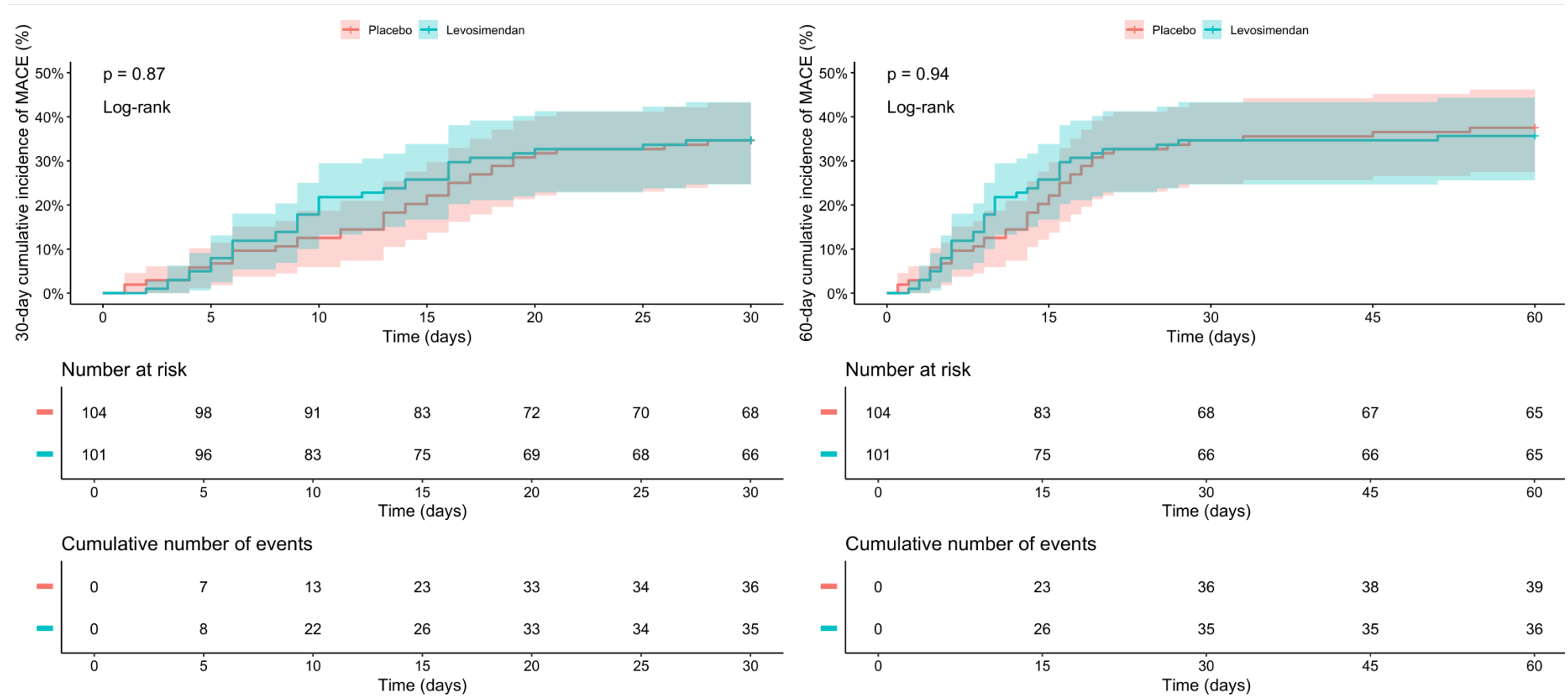
eFigure 2. Cumulative Incidence of D30 and D60 Mortality



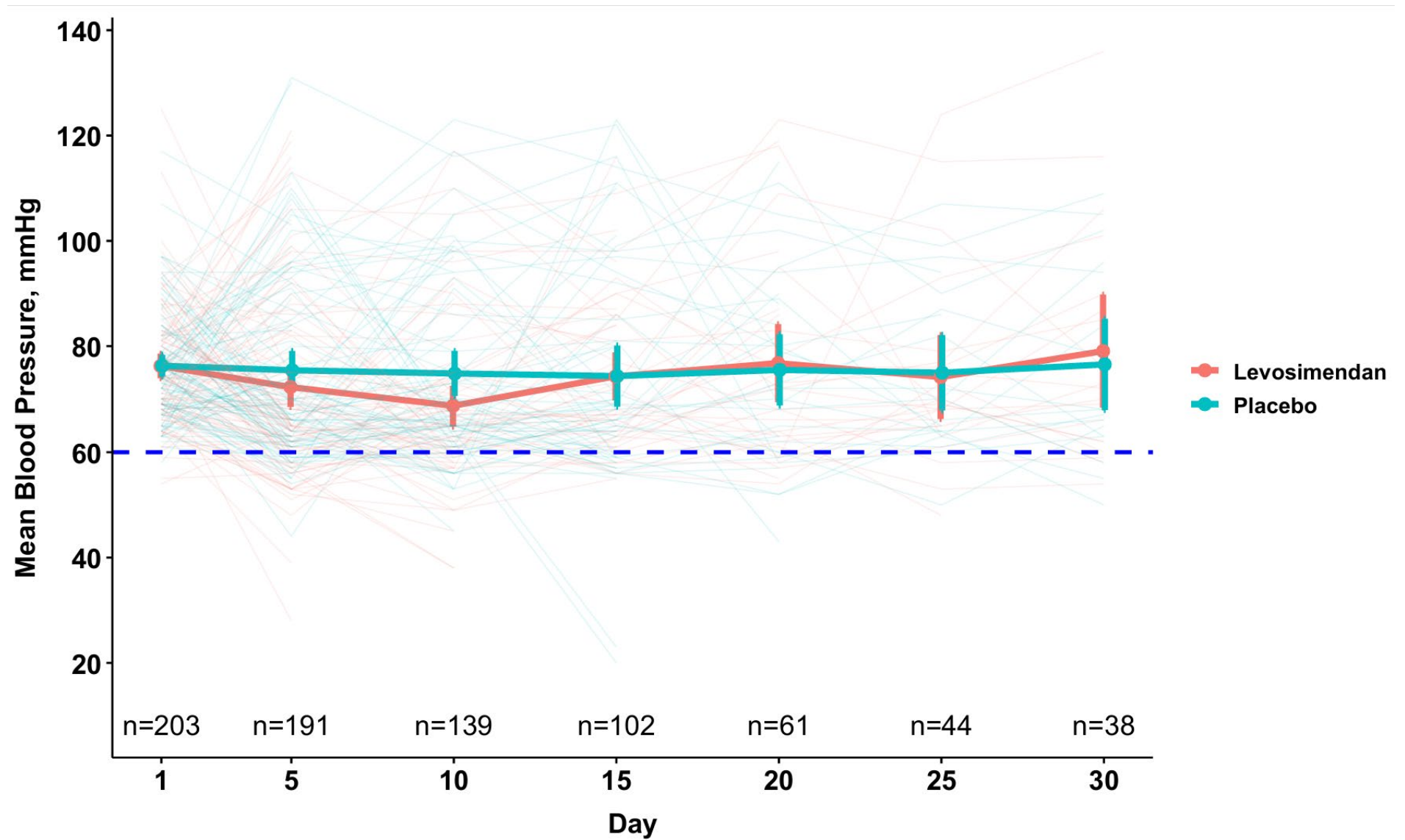
eFigure 3. SOFA Score Subcomponents Evolution From D1 to D28



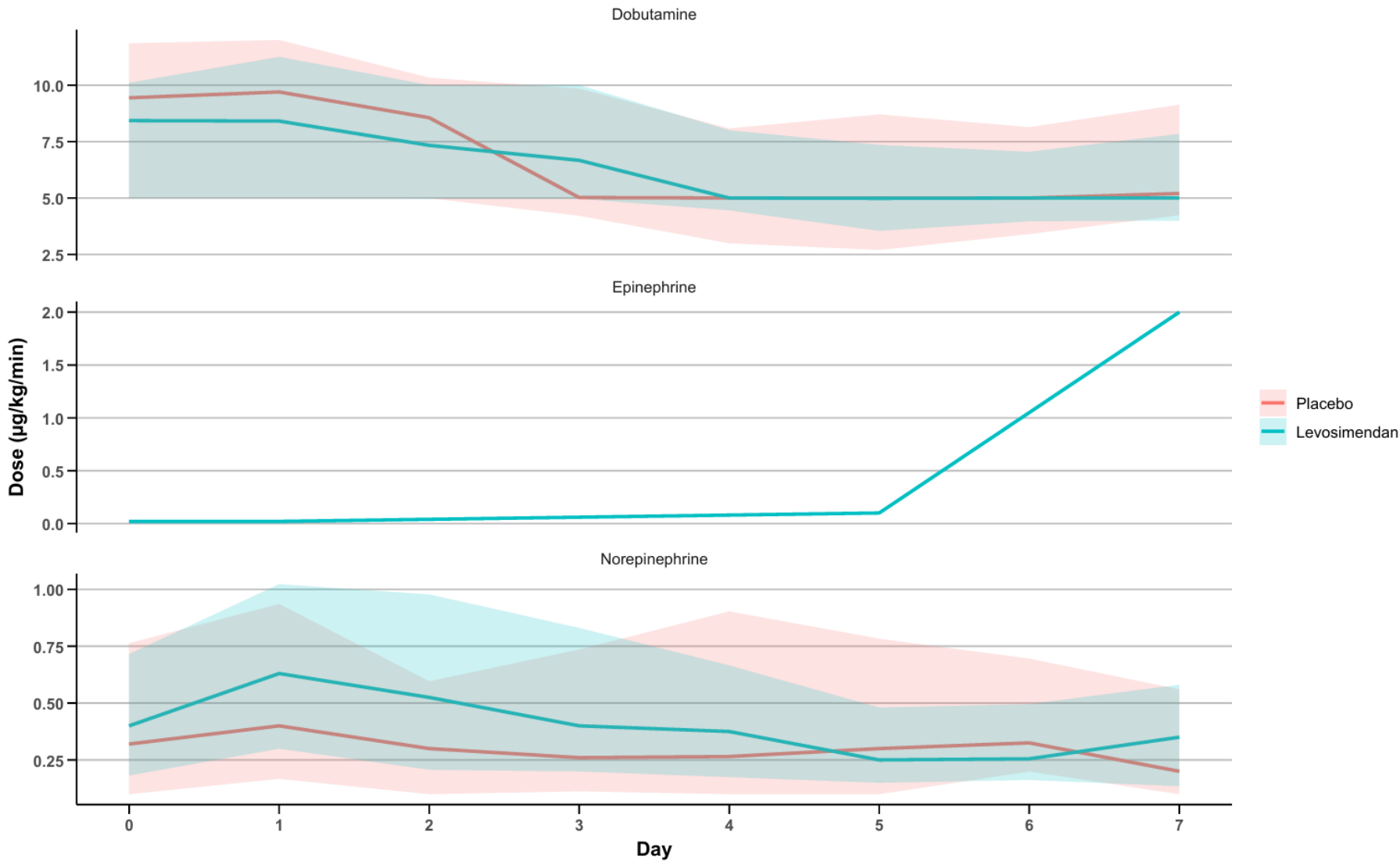
eFigure 4. D30 and D60 Cumulative Incidence of Major Cardiac Adverse Events (MACE) Defined as Death, Cardiac Transplant, Escalation to Permanent Left Ventricular Assist Device, Stroke, or Re-Hospitalization for Heart Failure



eFigure 5. Mean Blood Pressure Evolution From D1 to D30

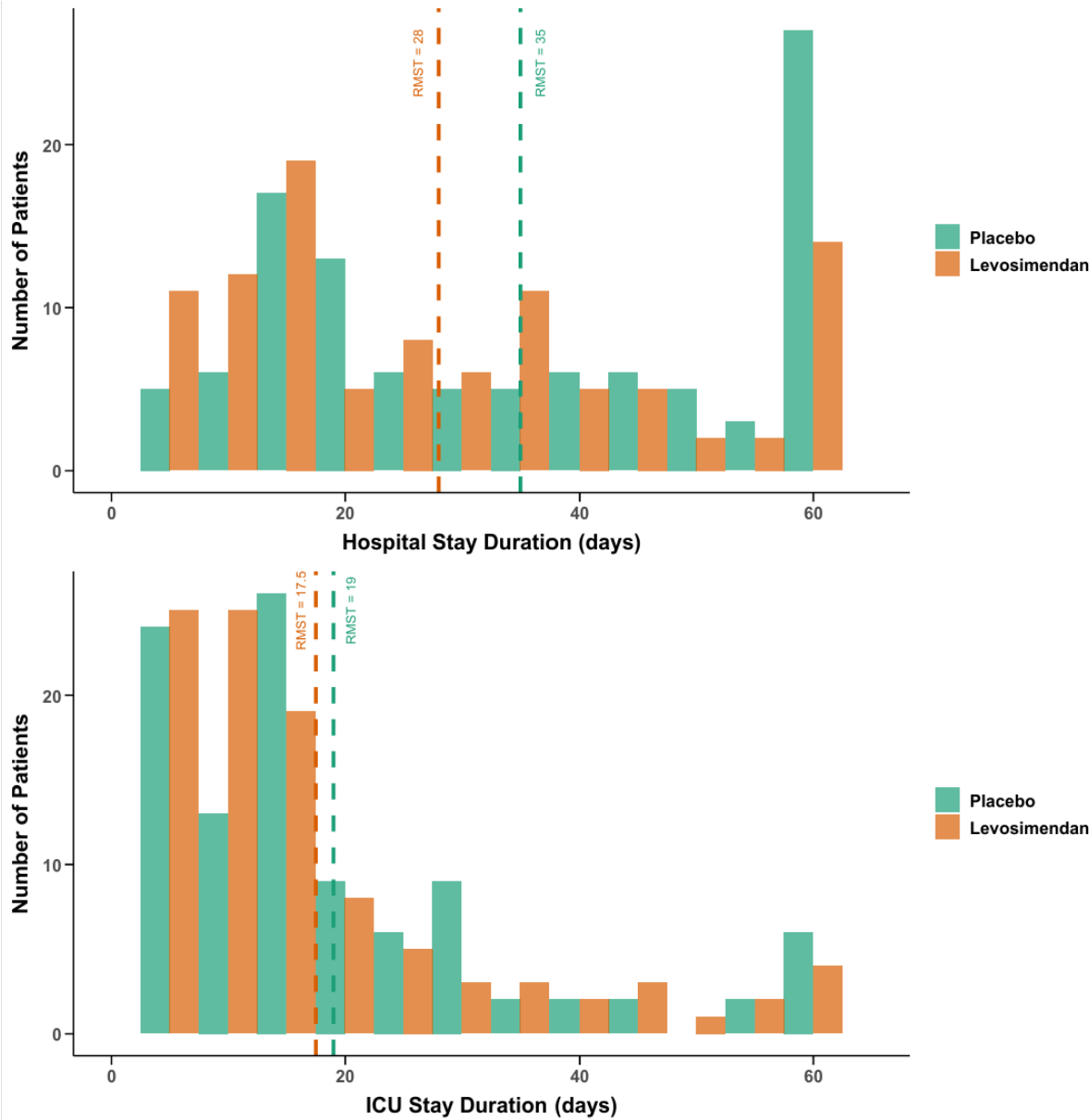


eFigure 6. Median (IQR) Doses of Vasoactive Drugs From Randomization (Day 0) to Day 7

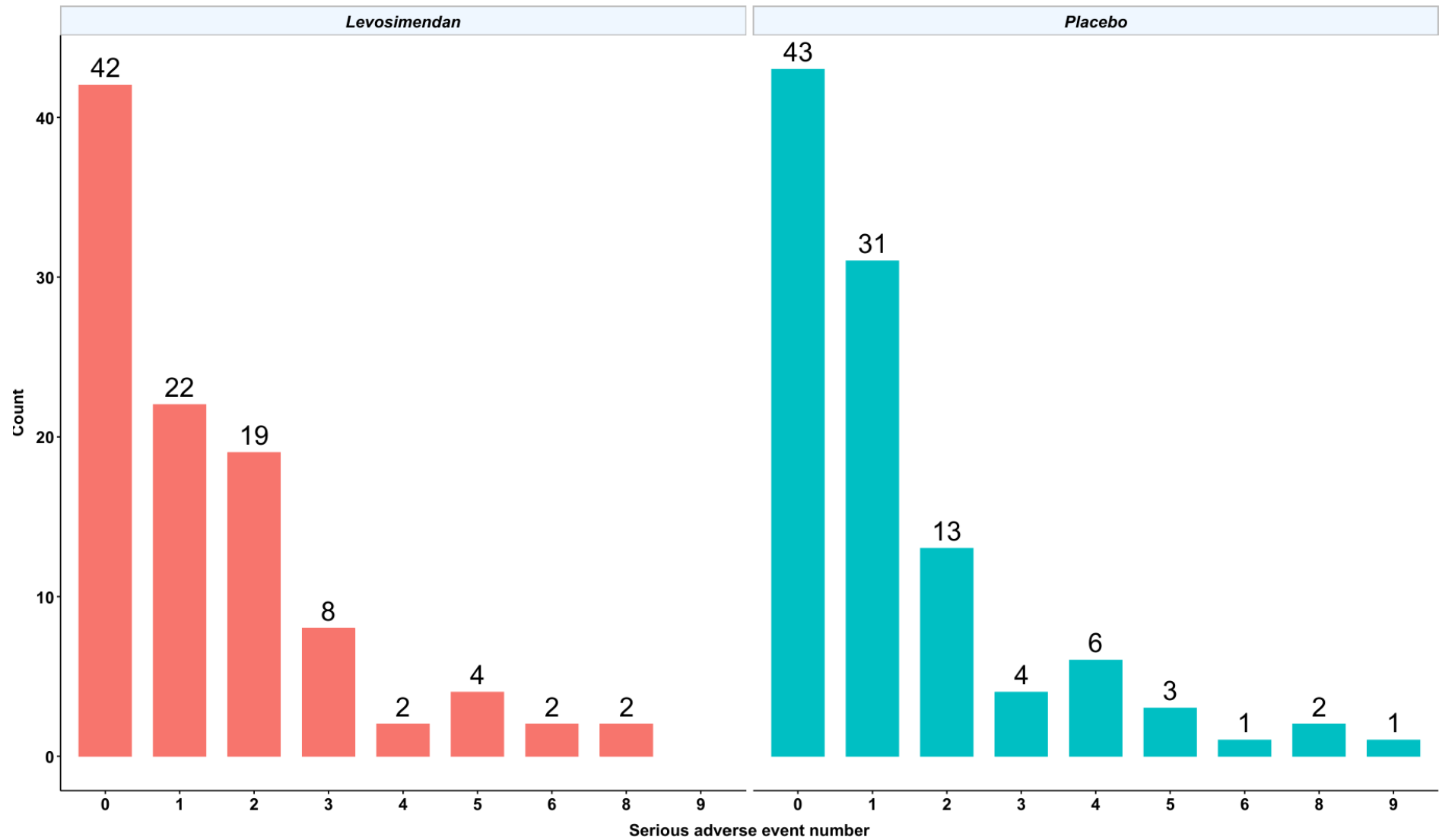


eFigure 7. Distribution of ICU and Hospital Stay Up to D60

RMST = Restricted Mean Survival Time



eFigure 8. Number of Serious Adverse Events



eFigure 9. Cumulative Incidence of Successful ECMO Weaning in Predefined Subgroups of Patients

